Singular Value Decomposition and Biplots

References

- Biplots in Practice. Michael Greenacre. Fundación BBVA, 2010.
- Multivariate Analysis of Ecological Data. Michael Greenacre, Raul Primicerio. Fundación BBVA, 2013.
- La práctica del Análisis de Correspondencias. Michael Greenacre. Fundación BBVA, 2008.
- Applied Multivariate Statistical Analysis. Johnson and Wichern.
- Modern Statistical Prediction and Machine Learning. Course by Gastón Sánchez.
- ▸ Link

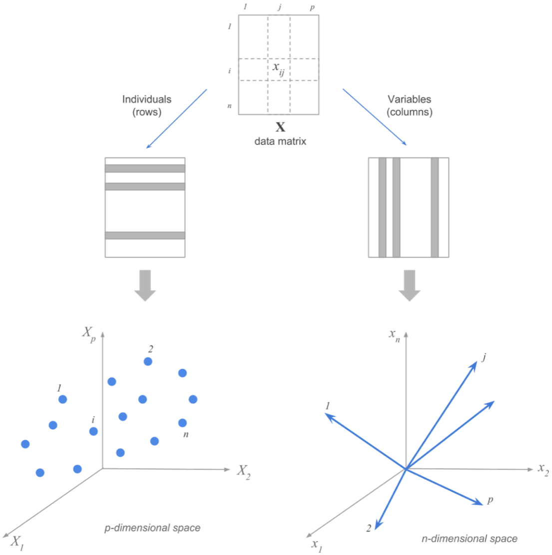
Data Matrix $X_{n\times p}$

$$X_{n\times p} = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{12} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{pmatrix} = [x_1, \cdots, x_p] = \begin{pmatrix} x_1' \\ \vdots \\ x_n' \end{pmatrix}$$

There are two different perspectives when looking at a data matrix:

- n observations in the rows
- p quantitative variables in the columns

Duality observations/variables



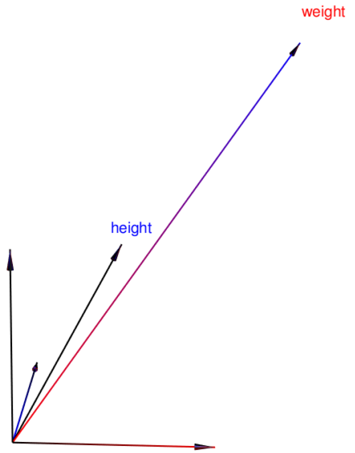
Duality: Toy example

3 observations, 2 variables

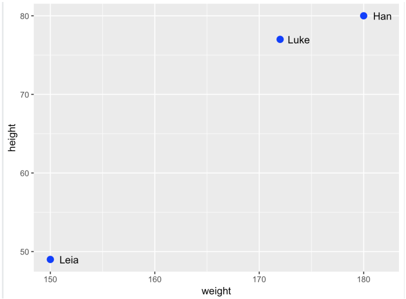
```
       weight height
Leia    150     49
Luke    172     77
Han     180     80
```

DUALITY: TOY EXAMPLE

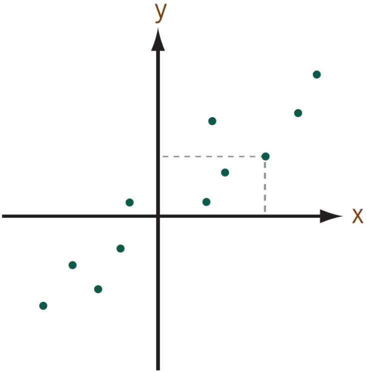
p vectors on an n-dimension space.



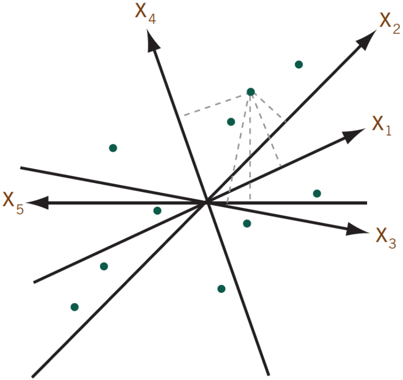
n points on a p-dimension space.



Scatterplot vs. Biplot



Scatterplot



Biplot

Biplot: a simple example

$$X_{5\times 4} = \begin{pmatrix} 8 & 2 & 2 & -6 \\ 5 & 0 & 3 & -4 \\ -2 & -3 & 3 & 1 \\ 2 & 3 & -3 & -1 \\ 4 & 6 & -6 & -2 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 1 & 2 \\ -1 & 1 \\ 1 & -1 \\ 2 & -2 \end{pmatrix} \begin{pmatrix} 3 & 2 & -1 & 2 \\ 1 & -1 & 2 & 1 \end{pmatrix}$$

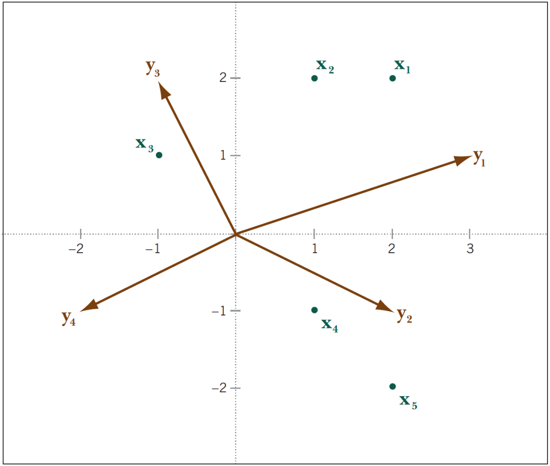
$$\text{target matrix} = \text{left matrix} \cdot \text{right matrix}$$

Biplot: a simple example

$$T = XY^T = \begin{pmatrix} x_1^T \\ x_2^T \\ x_3^T \\ x_4^T \\ x_5^T \end{pmatrix} \begin{pmatrix} y_1 & y_2 & y_3 & y_4 \end{pmatrix} = \begin{pmatrix} x_1^T y_1 & x_1^T y_2 & x_1^T y_3 & x_1^T y_4 \\ x_2^T y_1 & x_2^T y_2 & x_2^T y_3 & x_2^T y_4 \\ x_3^T y_1 & x_3^T y_2 & x_3^T y_3 & x_3^T y_4 \\ x_4^T y_1 & x_4^T y_2 & x_4^T y_3 & x_4^T y_4 \\ x_5^T y_1 & x_5^T y_2 & x_5^T y_3 & x_5^T y_4 \end{pmatrix}$$

Each element in the target matrix can be written as the scalar product of one vector of the left matrix and one vector of the right matrix.

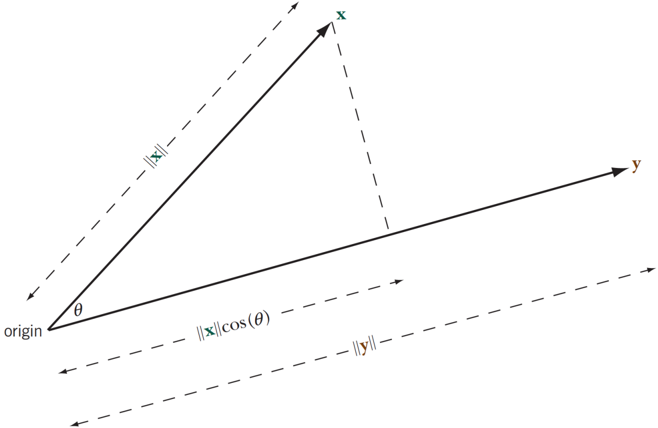
Biplot: a simple example



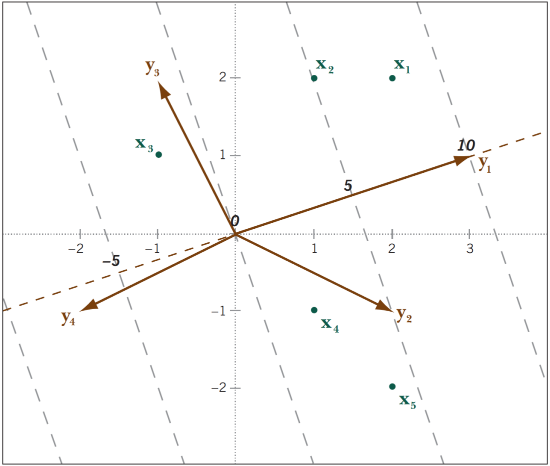
$$\mathbf{x}_1 = [\,2 \quad 2\,]^\mathsf{T}$$
$$\mathbf{x}_2 = [\,1 \quad 2\,]^\mathsf{T}$$
$$\mathbf{x}_3 = [-1 \quad 1\,]^\mathsf{T}$$
$$\mathbf{x}_4 = [\,1 \quad -1\,]^\mathsf{T}$$
$$\mathbf{x}_5 = [\,2 \quad -2\,]^\mathsf{T}$$

$$\mathbf{y}_1 = [\,3 \quad 1\,]^\mathsf{T}$$
$$\mathbf{y}_2 = [\,2 \quad -1\,]^\mathsf{T}$$
$$\mathbf{y}_3 = [-1 \quad 2\,]^\mathsf{T}$$
$$\mathbf{y}_4 = [-2 \quad -1\,]^\mathsf{T}$$

Scalar product. Geometrical interpretation.



Calibrating the biplot



Biplots: a powerful tool ...

- In a biplot, two sets of points are plotted together according to coordinates provided by the left and right matrices that result from the decomposition of a target matrix.
- With the convention that variables are represented as vectors and observations as points, one can read off the values in the target matrix by projecting each point onto the corresponding axis, once properly calibrated.
- It may be the case that the actual values are not as important as the ability to see how the cases line up along each variable.
- If two axes lie in the same orientation, we could infer that the cases have the same relative positioning along both variables and, consequently, high correlation between the variables.

Biplots: a powerful tool ...

Going back to slide 12, we see that

Variables y_1 and y_4 lie almost on the same axis, with opposite directions, meaning that high values for an observation on y_1 correspond to low values for the same observation on y_4. Therefore, a strong negative correlation between y_1 and y_4 can be expected.

$$cor(y_1, y_4) = -0.976$$

- For a given data matrix, we won't typically be able to find a decomposition into a product of matrices with only two or three dimensions: in fact, the number of dimensions needed is equal to the rank of the target matrix.

$$X_{5\times 4} = \begin{pmatrix} 8 & 2 & 2 & -6 \\ 5 & 0 & 3 & -4 \\ -2 & -3 & 3 & 1 \\ 2 & 3 & -3 & -1 \\ 4 & 6 & -6 & -2 \end{pmatrix}$$

This matrix has rank 2!! It contains redundant information.

… BUT

- For higher-dimensional situations we would need methods for reducing the dimensionality of the target matrix.
- Start talking about *Matrix Approximation*

Matrix Approximation

Suppose $X_{n \times m}$ is a matrix with rank r (typically $r = min\{n, m\}$). The idea is to find another $n \times m$ matrix $\hat{X}_{n \times m}$ of lower rank $p < r$ that resembles X as closely as possible.

Based on what criterion? Least-squares approximation.

We want to find a matrix $\hat{X}$ that minimizes the following objective function over all possible rank p matrices:

$$\text{trace}[(X - \hat{X})(X - \hat{X})^T] = \sum_{i=1}^{n} \sum_{j=1}^{m} (x_{ij} - \hat{x_{ij}})^2$$

As we can read on page 102 of Johnson and Wichern's book,

> The singular value decomposition is closely connected to a result concerning the approximation of a rectangular matrix by a lower-rank one, due to Eckart and Young.

Matrix Approximation

Eckart and Young's Theorem

Let A be an $n \times m$ matrix of real numbers with $n >> m$ and SVD equal to $U\Lambda V^T$. Let $p < r = rank(A)$. Then:

$$B = \sum_{i=1}^{p} \lambda_i \mathrm{u_i} \mathrm{v_i}^{\mathrm{T}}$$

is the rank-p least squares approximation to A. It minimizes

$$tr[(A-B)(A-B)^T] = \sum_{i=1}^{n} \sum_{j=1}^{m} (a_{ij} - b_{ij})^2$$

over all $n \times m$ matrices B having rank no greater than p. The minimum value, or error in the approximation, is $\sum_{i=p+1}^{r} \lambda_i^2$.

Matrix Approximation: Singular Value Decomposition

Let A be an $n \times m$ matrix of real numbers. Then there exist an $n \times n$ orthogonal matrix U and a $m \times m$ orthogonal matrix V such that:

$$A = U\Lambda V^T$$

where the $n \times m$ matrix Λ has (i, i) entry $\lambda_i \geq 0$ for $i = 1, 2, \ldots, \text{mín}(n, m)$ and the other entries are 0. The positive constants λ_i are called the *singular values* of A. The columns $\mathsf{u}_1, \mathsf{u}_2, \ldots, \mathsf{u}_n$ and $\mathsf{v}_1, \mathsf{v}_2, \ldots \mathsf{v}_m$ of U and V are called left and right singular vectors respectively.

This is the full version of the SVD.

Matrix Approximation: Singular Value Decomposition

Applied to our X matrix of rank r:

$$X_{n \times m} = U \Lambda V^T$$

where U is $n \times r$ (orthonormal by columns, $U^T U = I_r$), V is $m \times r$ (othonormal by columns, $V^T V = I_r$) and Λ is a $r \times r$ diagonal matrix with positive numbers $\lambda_1, \ldots, \lambda_r$ on the diagonal in descending order.

This decomposition provides the solution for the approximation problem in the exact way required for the biplot. The only decision is how to distribute the matrix Λ of singular values to the left and to the right in order to define the biplot's left matrix and right matrix.

2D Biplot

For a two-dimension biplot, if we take the *first two components* in matrices U, V and Λ, we can write:

$$\hat{X} = U_{[2]}\Lambda_{[2]}V_{[2]}^T$$

$\hat{X}$ is the best 2-rank approximation matrix to our original X and its SVD provides the elements to graph the biplot. If we join the first two matrices, we obtain a decomposition as the product of a left matrix and a right matrix:

$$\underbrace{\hat{X}}_{n\times m} = \underbrace{Z_{[2]}}_{n\times 2}\underbrace{V_{[2]}^T}_{2\times m}$$

We can say that the information contained in $n \times m$ values is compressed into $n \times 2$ values.

2D Biplot: quality of the representation

The singular values provide a quantification of the closeness of the approximation of $\hat{X}$ to X. In the case of the 2D Biplot:

$$\frac{\lambda_1^2 + \lambda_2^2}{\sum_{i=1}^{r} \lambda_i^2}$$

The quality of the representation depends on the magnitude of the sum of the first two squared singular values compared to the sum of all squared singular values, different from 0. Expressed as a percentage, it is the *amount of variability of the original data matrix taken into account by this lower-dimension representation*.

What if we don't take 2 components but, let's say, p?

The quality of the representation is going to be greater, but for $p > 3$ we are not going to be able to see a plot of the resulting reduced data. You cannot display or imagine the reduced configuration.

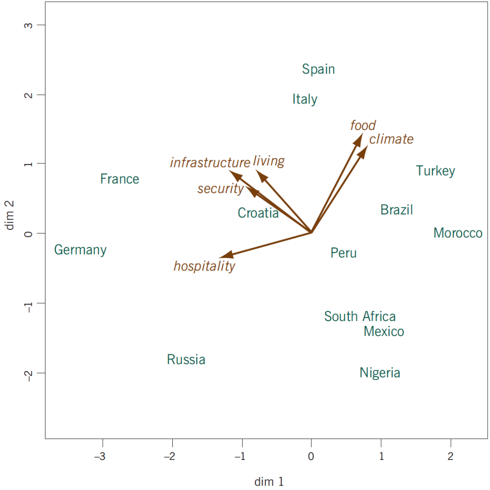
An Example

Data: ratings of 13 countries on six attributes: standard of living, climate, food, security, hospitality and infrastructure.

Countries	*living*	*climate*	*food*	*security*	*hospitality*	*infrastructure*
Italy	7	8	9	5	3	7
Spain	7	9	9	5	2	8
Croatia	5	6	6	6	5	6
Brazil	5	8	7	3	2	3
Russia	6	2	2	3	7	6
Germany	8	3	2	8	7	9
Turkey	5	8	9	3	1	3
Morocco	4	7	8	2	1	2
Peru	5	6	6	3	4	4
Nigeria	2	4	4	2	3	2
France	8	4	7	7	9	8
Mexico	2	5	5	2	3	3
South Africa	4	4	5	3	3	3

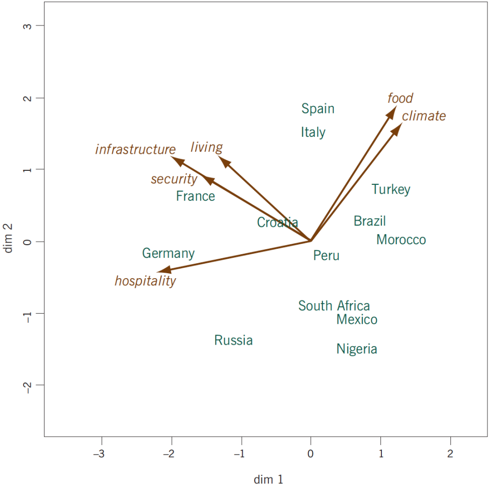
An Example

Form biplot or row metric preserving biplot: singular values are assigned to the left matrix.



An Example

Covariance biplot or column metric preserving biplot: singular values are assigned to the right matrix.



The power of the SVD

- It is the essence in the biplot generation. It is the building block in most of the multivariate techniques.
- Different Biplots for different data types: as long as the matrix to be approximated can be appropriately set up according to the nature of the data:
  - Principal Component Analysis Biplots
  - Correspondence Analysis Biplots and Multiple Correspondence Analysis Biplots
  - Discriminant Analysis Biplot
  - Multidimensional Scaling Biplots
  - Log-Ratio Biplots
  - Nonlinear Biplots